Text-Driven Adaptive Semantic Alignment Network for Cross-Scene Hyperspectral Image Classification

Wenzhen Wang, Fang Liu, *Member, IEEE*, Hongyuan Zhu, *Member, IEEE*, and Liang Xiao, *Senior Member, IEEE*

***Abstract*— Land cover in different scenes generally exhibits scene-invariant category semantic, typically represented and described consistently in a textual modality. Traditional cross-scene classification methods often treat categories as discrete class labels, neglecting their semantic information, or use category names merely as auxiliary textual modalities to enhance the discriminative representations of land cover. However, the cross-scene consistency of category semantic for land cover remains underexplored and underutilized. To address this issue, the text-driven adaptive semantic alignment network (TASA-Net) is proposed in this article for cross-scene hyperspectral image classification (HSIC). TASA-Net employs hand-crafted template prompts for stable category descriptions and vision-guided fine semantic prompts (VG-FSPs) for dynamic scene adaptation. Through a dual-gated adaptive mechanism, TASA-Net optimally weights coarse- and fine-grained semantics in a shared space, ensuring stable yet discriminative semantic representation. Additionally, cross-modal semantic alignment projects visual features into the shared semantic space, while a soft alignment strategy dynamically adjusts category correlations to enhance intraclass consistency and mitigate domain shifts. Ultimately, by leveraging text-driven semantic consistency representation, TASA-Net achieves zero-shot cross-scene transfer for unsupervised classification. Experiments demonstrate superior performance across multiple hyperspectral datasets, validating the critical role of textual modality in enhancing model robustness and cross-scene generalization ability.**

Received 30 December 2024; revised 13 February 2025; accepted 3 March 2025. Date of publication 6 March 2025; date of current version 1 April 2025. This work was supported in part by the Frontier Technologies Research and Development Program of Jiangsu under Grant BF2024070, in part by Jiangsu Geological Bureau Research Project under Grant 2023KY11, in part by the National Natural Science Foundation of China under Grant 62471235, in part by China Scholarship Council, and in part by the Economic Development Board (EDB) Space Technology Development Grant Program of Singapore under Grant S22-19016-STDP. *(Corresponding author: Liang Xiao.)*

Wenzhen Wang is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: wangwz@njust.edu.cn).

Fang Liu is with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing 210094, China, also with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan 430072, China, and also with Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology, Nanjing 210095, China (e-mail: liufang_cs@njust.edu.cn).

Hongyuan Zhu is with the Institute for Infocomm Research (I^2R), Agency for Science, Technology, and Research (A*STAR), Singapore 138632 (e-mail: zhuh@i2r.a-star.edu.sg).

Liang Xiao is with the School of Computer Science and Engineering, Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, Nanjing University of Science and Technology, Nanjing 210094, China (e-mail: xiaoliang@mail.njust.edu.cn).

Digital Object Identifier 10.1109/TGRS.2025.3548607

***Index Terms*— Cross-modal semantic alignment, cross-scene classification, semantic descriptions, textual modality.**

I. Introduction

CROSS-SCENE hyperspectral image classification (HSIC) aims to train a model with strong generalization capabilities using hyperspectral data and class labels from the source scene, enabling accurate classification in the target scene where data share consistent classes but exhibit different distributions. The core challenge lies in the diversity of data distributions and significant domain discrepancies between scenes [1]. Traditional single-modal visual methods often struggle to maintain stable classification performance when confronted with scene variation and category distribution shifts. This is because visual information from hyperspectral images may be affected by factors such as lighting, environment, and resolution, causing significant deviations in visual representations and consequently reducing model generalization performance. In contrast, the textual modality exhibits relatively stable semantic characteristics that are less influenced by changes in scene conditions [2]. It can provide additional semantic constraints for the visual modality, enhancing the abilities of the model to understand and consistently represent land cover across different scenes. Therefore, incorporating the textual modality offers significant advantages for cross-scene HSIC.

The integration of textual modality with hyperspectral visual images in multimodal learning has shown promising results in various studies [3], [4], [5]. These studies mainly address two critical questions as follows.

1) *WHAT:* What is the appropriate formulation of the concept space corresponding to the textual modality for HSIC tasks?
2) *HOW:* How can the textual modality effectively inject its semantic information into the visual modality?

*To address the **first challenge** of effective concept space formulation:* Textual modality primarily manifests as class labels (e.g., "playground," "buildings," and "trees") in land cover classification, which carry fundamental semantic information about land cover objects. However, in traditional visual classification methods, these textual labels are often simply converted to discrete labels [as shown in Fig. 1(a)] and used solely for cross-entropy loss computation in class prediction [6]. While this approach offers some utility in

1558-0644 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.
Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

classification tasks, it neglects much of the latent semantic information contained within the class labels, limiting the ability of the model to understand and generalize deep semantic concepts.

Inspired by pretrained vision-language models, multimodal learning that integrates textual modality has increasingly become a research hotspot in the field of HSIC. As shown in Fig. 1(b), many studies combine class labels with prompt templates to serve as text inputs, providing stable semantic structure. For example, "A hyperspectral image of [CLASS]." is used as a template, where [CLASS] represents the specific class label [7]. This approach offers a unified semantic background for the visual modality, aiding the model in better understanding category characteristics. However, due to the coarse-grained nature of the templates, it is challenging to capture the deep semantic information of land cover. Consequently, some researchers propose to optimize the semantic space of categories using more fine-grained, hand-crafted text descriptions [8]. By adding detailed semantic descriptions for each category, such as "The [playground] has a red running track." for the "playground" label, models can gain richer semantic information and improve their ability to adapt to different scenes.

Different text descriptions can represent the same hyperspectral image from various perspectives, potentially emphasizing different characteristics or environmental backgrounds of land cover, such as shape, color, or spatial position. While diverse, hand-crafted text descriptions enrich the semantics information of land cover objects, they also introduce substantial design costs and subjectivity. To address this, recent research explores adaptive prompt generation methods [9], [10], as shown in Fig. 1(c). Using parameter-learning models, these methods automatically generate text descriptions tailored to specific tasks, dynamically creating text inputs that fit task demands. This avoids the subjectivity and limitations of hand-crafted designs, producing task-relevant descriptions that enhance the synergy between textual and visual modalities and improve model performance in complex scenarios. Nevertheless, hyperspectral images often contain more semantic richness than their corresponding textual class labels due to their high-dimensional and complex nature. Moreover, the same land cover category may exhibit different spectral and spatial characteristics across scenes, making it challenging for a simple class-labels-generated semantic space to accommodate these variations and reduce generalization capability.

To address the ***second challenge*** *of effective visual-text fusion:* Multimodal learning methods of textual data and hyperspectral images can be summarized into three paradigms based on how textual information acts on visual representations: modality fusion, modality interaction, and modality alignment. The modality fusion paradigm, which is the most common and widely applied, uses textual modality as auxiliary information to enrich the visual representations of hyperspectral images. By combining features through concatenation, weighted averaging, or attention mechanisms, both modalities create joint representations that enhance multimodal understanding of the land cover objects [11]. For instance, the

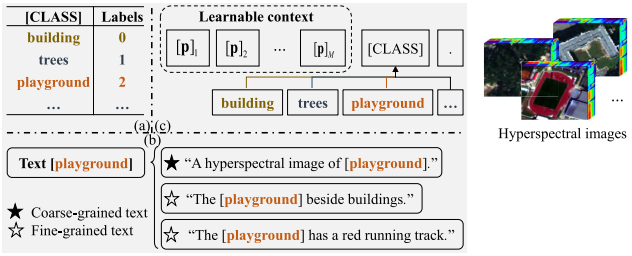


Fig. 1. Types of textual modality corresponding to hyperspectral images. (a) Discrete Labeling. Textual class labels are simply converted to discrete labels for cross-entropy loss computation in class prediction. (b) Hand-crafted prompt templates. Class labels are combined with prompt templates to provide a stable text structure. (c) Parametric learned prompts. Semantic descriptions are automatically generated according to specific tasks using a parameter-learning approach to adaptively create task-oriented textual modality inputs.

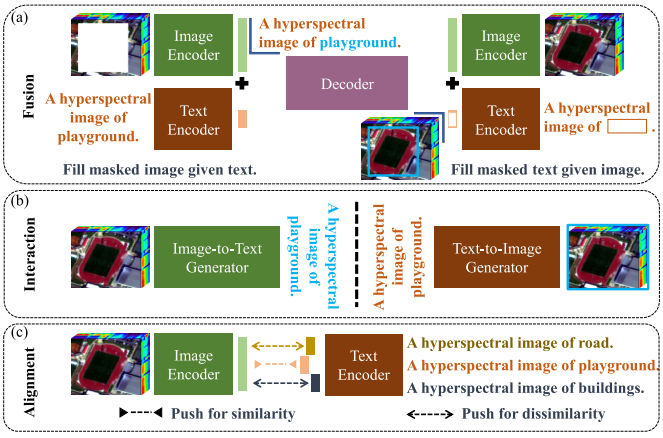


Fig. 2. Three multimodal learning paradigms for textual and hyperspectral visual images collaboration. (a) Modality fusion paradigm. The textual modality primarily plays a passive auxiliary role, enriching visual representations. (b) Modality interaction paradigm. The textual modality acts as a deeply interactive participant and guide, leading visual information decoding or text generation. (c) Modality alignment paradigm. The textual modality serves as an active aligning force, aligning visual representations for semantic consistency. Textual modality exhibits shared capabilities across these paradigms to enrich visual information, enhance model robustness, and utilize semantic consistency.

FUS-MAE method, as shown in Fig. 2(a), applies masking operations to hyperspectral images (text labels) for joint data reconstruction to achieve deep multimodal data fusion [12]. This paradigm mainly aims to improve image classification and object detection performance but may not thoroughly address deep-rooted differences between modalities, especially when there are significant scene changes.

The modality interaction paradigm facilitates dynamic information exchange and transformation between text and visual images through cross-attention mechanisms or generative models. Here, the textual modality not only enriches the visual representation but also guides visual decoding or text generation, widely applied in tasks like image captioning [13] and visual question answering [14]. For example, the Diffusion-Geo method, as illustrated in Fig. 2(b), transforms text descriptions into hyperspectral images with specified semantics through a generative model, establishing deeper interactions in cross-modality tasks [15]. Although this paradigm performs well in generative tasks, its direct contribution to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

classification tasks is relatively limited, especially in handling domain shifts across scenes.

In the modality alignment paradigm, textual modality aligns the visual representations of hyperspectral images, aiming to map both modalities into a shared representation space for semantic alignment. Common methods include contrastive learning [16] or nearest neighbor clustering [17], improving cross-modality tasks such as image retrieval or text-image matching. For instance, RemoteCLIP, shown in Fig. 2(c), aligns image and corresponding text in a shared embedding space through contrastive learning, enhancing cross-modal consistency and interpretability [18]. This paradigm is particularly effective for cross-scene tasks as it directly addresses the inherent inconsistency in visual representations across scenes. It reduces the negative impact of domain differences while using the stable semantic features of textual modality as anchors to guide consistent representations of land cover objects across different scenes. However, focusing solely on aligning fine-grained text descriptions with visual images can limit the ability of the model to generalize to diverse text descriptions. Therefore, models based on this paradigm must balance text description diversity and consistency while ensuring generalization capability for cross-scene tasks to enhance adaptability and robustness.

Therefore, our research focuses on developing robust semantic representations using the textual modality to address key challenges, such as data distribution diversity and significant domain differences in cross-scene HSIC. To achieve this, a text-driven adaptive semantic alignment network (TASA-Net) is specifically proposed in this article. The network first incorporates the feature diversity of the same category across different scenes into the semantic space construction, generating fine-grained semantic descriptions based on spectral features, spatial relationships, and other visual contexts. This approach not only captures complex content in hyperspectral images dynamically but also provides more robust generalization and consistency for cross-scene land cover representations. Second, the proposed TASA-Net employs a dual-gated adaptive strategy to integrate fine-grained semantics guided by visual features with coarse-grained semantics generated by hand-crafted templates, enhancing both the consistency and discriminability of category representations across different scenes. Finally, the network achieves zero-shot transfer classification in the unseen target scenario through a cross-modal semantic alignment mechanism in a shared space. The main contributions are as follows.

1) A vision-guided fine semantic prompt (VG-FSP) module is designed to mine fine-grained semantics from key visual representations of hyperspectral images, dynamically generating scene-adaptive category descriptions. This significantly enhances the adaptability and discriminative capability of semantic representations in different scenes.
2) A dual-gated adaptive semantic representation (DG-ASR) module is proposed, along with a dual-constrained dynamic fusion strategy that adaptively adjusts the weights of fine-grained semantics from visual prompts and coarse-grained semantics from textual prompts, thereby achieving robust semantic representations under cross-scene variations.
3) By aligning visual and textual modalities in a shared semantic space, this work generates adaptive semantic weights for categories, supporting zero-shot transfer tasks across scenes without requiring explicit classifiers or domain adaptation optimizations.

The remainder of this article is organized as follows. Section II reviews the related work. Section III provides a detailed description of the proposed method. Experimental results are presented in Section IV, followed by an in-depth discussion in Section V. Finally, Section VI concludes the article.

II. Related Work

A. Cross-Scene Image Classification

Cross-scene classification methods have achieved significant progress in the fields of medical imaging, natural image analysis, and multispectral remote sensing. In medical imaging, Zhu et al. [19] alleviate performance degradation caused by imaging differences across medical devices through an anatomical structure-constrained domain generalization approach. Such methods assume cross-scene discrepancies can be modeled via low-dimensional linear transformations (e.g., illumination adjustments) or structural consistency (e.g., organ shapes). For natural images, researchers address scene variations through domain-invariant feature alignment. Long et al. [20] propose a maximum mean discrepancy (MMD)-based deep domain adaptation framework to minimize feature distribution differences between source and target domains, while Bi et al. [21] employ adversarial training to learn domain-invariant representations. However, domain alignment methods [22], [23], [24], [25], [26], relying on data-driven brute-force fitting, show limited effectiveness in hyperspectral small-sample scenarios. Recent advances in vision-language multimodal approaches offer new insight for cross-scene remote sensing image classification. Zhang et al. [8] propose to align visual features to a shared semantic space, leveraging the cross-scene stability of text labels for zero-shot transfer. Singha et al. [9] further introduce textual descriptions to guide semantic understanding of land cover in new scenes. Dang et al. [27] enhance cross-scene generalization by integrating geographic semantic priors, such as "vegetation exhibits high near-infrared reflectance." However, these methods depend on fixed text templates and struggle to adapt to nonlinear spectral shifts in hyperspectral imagery.

The core challenge in cross-scene HSIC stems from its physical properties: the spectral reflectance of materials undergoes nonlinear shifts due to environmental variations. For instance, vegetation in humid environments may exhibit shifted water absorption peaks in shortwave infrared bands, which fixed text templates fail to capture. The proposed method transcends traditional static alignment by introducing a dynamic semantic adaptation mechanism that injects scene-specific spectral statistics into text description

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

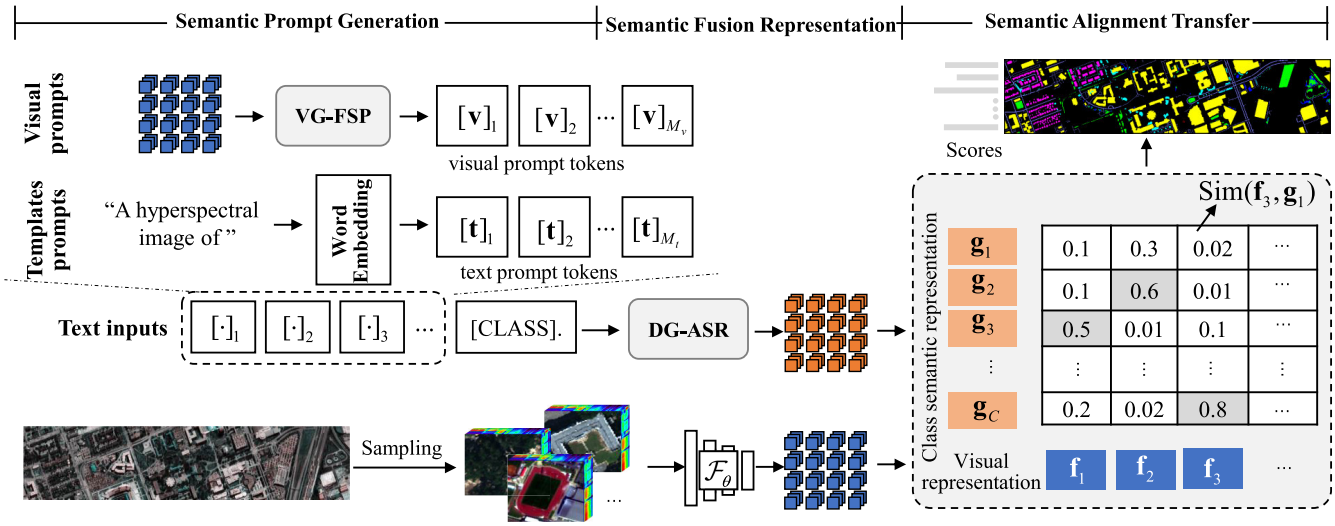


Fig. 3. Flowchart of TASA-Net. It includes three parts: Semantic prompt generation, semantic fusion representation, and semantic alignment transfer. The VG-FSP module designed in semantic prompt generation dynamically extracts fine-grained, scene-adaptive category descriptions from hyperspectral visual features, enhancing semantic adaptability. The DG-ASR module proposed in semantic fusion representation employs a dual-constrained dynamic fusion strategy to balance fine-grained visual semantics and coarse-grained textual semantics, ensuring robust representations under cross-scene variations. Together, these modules enable effective semantic alignment and transfer for improved classification performance.

generation, enabling environment-aware semantic evolution while establishing cross-scene invariance in semantic space.

B. Multimodal Semantic Representation

In cross-scene land cover classification, hyperspectral imagery achieves precise characterization through spectral fingerprint features derived from hundreds of contiguous spectral bands, complemented by spatial texture information. Traditional methods extract visual features via hierarchical convolutions of CNNs or self-attention mechanisms of ViT, yet such unimodal approaches lack explicit cross-scene semantic consistency. Recent studies propose domain-specific architectures for feature optimization. For example, DiffDet4SAR [28] enhances target feature robustness in SAR imagery using diffusion models, while HiEUM [29] achieves hierarchical spatial decoupling via sparse convolutions. However, despite improving feature modeling, these methods lack explicit semantic guidance for cross-scene stability.

Vision-language pretraining (VLP) models like CLIP [30] and BLIP [31] align visual features with textual semantics in shared embedding spaces via contrastive learning, enabling semantic constraints for visual representation. Their strength lies in leveraging interpretable semantic concepts (e.g., material composition and spatial layouts) to establish cross-scene invariance. For instance, describing "urban areas contain dense grid-like road networks" guides road texture recognition under varying illumination. However, existing VLP models face limitations when applied to cross-scene HSIC, including the rigidity of fixed text templates and insufficient fine-grained representation capabilities. First, mainstream VLP models predominantly employ fixed textual prompt templates for vision-language alignment, such as CLIP's "A photo of a [CLASS]," which fails to adequately capture the spectral characteristics of different land cover categories in hyperspectral data. Second, hyperspectral classification often hinges on subtle spectral variations, yet existing VLP frameworks are primarily trained on RGB images and lack spectral modeling capabilities, restricting their ability to fully exploit the high-dimensional features of hyperspectral data. Additionally, hyperspectral classification frequently involves fine-grained land cover differentiation, such as distinguishing between different crop or soil types, a task that remains challenging for current VLP approaches.

III. Proposed Method

In closed-set cross-scene HSIC research, the target and source scenes share the same category text $\mathcal{T} = \{t_c\}_{c=1}^{C}$, where C denotes the number of categories and t_c represents the text label form of each category c. Our work constructs a training dataset $\mathcal{D} = \{\mathcal{X}, \mathcal{T}, \mathcal{Y}\}$ based on trainable hyperspectral visual image patches $\mathcal{X} = \{\mathbf{X}_i\}_{i=1}^{N}$ in the source scene and their corresponding discrete class labels $\mathcal{Y} = \{y_i\}_{i=1}^{N}$. Here, N represents the number of trainable samples, $\mathbf{X}_i \in \mathbb{R}^{s\times s\times d}$ is the ith image patch, s and d denote its spatial size and the spectral bands respectively. The flowchart of the proposed TASA-Net is shown in Fig. 3, comprising three parts: semantic prompt generation, semantic fusion representation, and semantic alignment transfer. First, in the semantic prompt generation part, a unified hand-crafted prompt template and adaptively generated contextual visual prompt optimize the original category semantic description, enhancing the stability and specificity of category text input. Next, in the semantic fusion representation part, the category semantic descriptions of generated text and visual prompts are integrated in a dual-gated adaptive learning manner, enriching category semantic representation. Finally, the semantic alignment transfer part aligns the semantic representations of visual images and category texts, achieving transfer and classification with semantic weighting.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

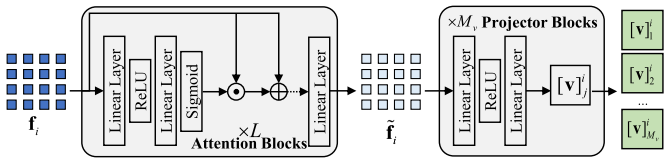


Fig. 4. Architecture of the designed VG-FSP. The hyperspectral image features are extracted through multiple attention layers to capture key contextual information. Subsequently, multiple projector blocks generate visual prompt tokens, providing more instructive visual contextual information.

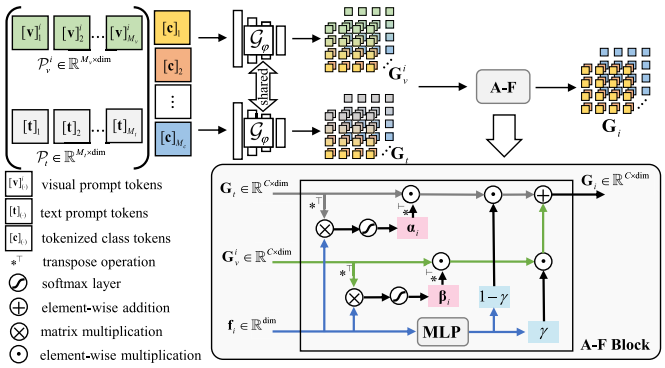


Fig. 5. Architecture of the designed DG-ASR. This module employs a dual-gated adaptive mechanism to efficiently fuse visual prompts with category text prompts. The figure illustrates the prompt-based semantic encoding process and the dual-gated adaptive fusion mechanism, which dynamically adjusts the weighting between fine-grained and coarse-grained semantics to improve the robustness and accuracy of cross-scene land cover classification.

A. Semantic Prompt Generation

In the semantic prompt generation part, the spectral characteristics, spatial relationships, and other visual contextual information of hyperspectral images are further mined to generate fine semantic descriptions for corresponding class labels. Additionally, a unified hand-crafted prompt template is added to the rough semantic description of the category text to enhance the stability of text input.

1) Vision Prompts: In this section, the VG-FSP module shown in Fig. 4 is proposed. It takes the visual representation $\mathbf{f}_i$ of the ith hyperspectral image patch $\mathbf{X}_i \in \mathcal{X}$ as input, where $\mathbf{f}_i = \mathcal{F}_\theta(\mathbf{X}_i) \in \mathbb{R}^{1\times \mathrm{dim}}$, $\mathcal{F}_\theta$ is the visual encoder parameterized by θ and dim is the feature dimension of the flattened feature maps. Here we affirm that the symbol i refers exclusively to the sample data and parameters associated with the ith image patch. Then, L attention blocks are designed for visual attention representation, mining critical contextual information of the visual image. For each layer $l (l = 1, 2, \ldots, L)$, the attention block is calculated as follows:

$$\boldsymbol{\omega}_i^{(l)} = \sigma\left(\mathbf{W}_2^{(l)} \cdot \mathrm{ReLU}\left(\mathbf{W}_1^{(l)}\mathbf{f}_i^{(l-1)} + \mathbf{b}_1^{(l)}\right) + \mathbf{b}_2^{(l)}\right). \quad (1)$$

Here, $\mathbf{W}_1^{(l)}, \mathbf{W}_2^{(l)}, \mathbf{b}_1^{(l)}, \mathbf{b}_2^{(l)}$ denote the linear layer parameters of the lth layer, σ is the sigmoid activation function, and $\mathbf{f}_i^{(0)} = \mathbf{f}_i$ is the input feature of the first attention block. The generated attention weights $\boldsymbol{\omega}_i^{(l)}$ are used to weight the input features, with residual connections introduced to preserve the original feature information, yielding the output feature for each layer

$$\mathbf{f}_i^{(l)} = \mathbf{f}_i^{(l-1)} + \boldsymbol{\omega}_i^{(l)} \odot \mathbf{f}_i^{(l-1)} \quad (2)$$

where $\odot$ represents element-wise multiplication. This cascade residual mechanism enhances the multilevel feature representation, enabling the model to progressively extract higher level details and contextual information while preserving original information. After L layers of attention blocks, the final visual attention representation is given by

$$\tilde{\mathbf{f}}_i = \mathbf{f}_i^{(L)}. \quad (3)$$

Subsequently, $\tilde{\mathbf{f}}_i$ is sent to M_v projector blocks to generate M_v visual prompt tokens

$$\mathcal{P}_v^i = \left\{[\mathbf{v}]_1^i, [\mathbf{v}]_2^i, \ldots, [\mathbf{v}]_{M_v}^i\right\} \quad (4)$$

where each projector block consists of a linear layer, a ReLU activation layer, and another linear layer. Each visual prompt token can be represented as

$$[\mathbf{v}]_j^i = \mathbf{W}_j^{(2)} \cdot \mathrm{ReLU}\left(\mathbf{W}_j^{(1)}\tilde{\mathbf{f}}_i + \mathbf{b}_j^{(1)}\right) + \mathbf{b}_j^{(2)}. \quad (5)$$

$\mathbf{W}_j^{(1)}, \mathbf{W}_j^{(2)}, \mathbf{b}_j^{(1)}, \mathbf{b}_j^{(2)}$ denote the linear layer parameters of the jth projector block. $[\mathbf{v}]_j^i \in \mathbb{R}^{\mathrm{dim}} (j = 1, 2, \ldots, M_v)$ contains fine semantic description information extracted from the hyperspectral image patch $\mathbf{X}_i$, dynamically generating more instructive visual context prompts for category text.

2) Templates Prompts: Although visual prompts provide refined category representation information, textual template prompts remain essential. Hand-crafted templates establish a consistent text structure for each category, enhancing semantic understanding consistency and offering a stable text framework that maintains better generalization across scenes and sample variations. This dual-prompt approach further optimizes language–image alignment, ensuring classification performance robustness and accuracy. Text template prompt tokens are represented as $\mathcal{P}_t = \{[\mathbf{t}]_1, [\mathbf{t}]_2, \ldots, [\mathbf{t}]_{M_t}\}$, where each $[\mathbf{t}]_k \in \mathbb{R}^{\mathrm{dim}} (k = 1, 2, \ldots, M_t)$ is derived from tokenized prompt words.

B. Semantic Fusion Representation

To achieve efficient integration of visual contextual prompts and textual template prompts, a DG-ASR module is designed to enhance the discriminative and generalizable representation of category semantics. In this module, a dual-constrained dynamic fusion strategy adaptively adjusts the weights of fine-grained semantics from visual contextual information and coarse-grained semantics from textual prompts, achieving robust semantic representations under cross-scene variations.

1) Prompt-Based Semantic Encoding: As shown in Fig. 5, for a hyperspectral image patch $\mathbf{X}_i \in \mathcal{X}$, its visual contextual prompt tokens $\mathcal{P}_v^i$ are concatenated in front of the tokenized class tokens $\mathcal{P}_c = \{[\mathbf{c}]_1, [\mathbf{c}]_2, \ldots, [\mathbf{c}]_{M_c}\}, (M_c = C)$. The visual-prompted semantic input for C categories can be expressed as $\mathcal{T}_v^i = \{\mathcal{P}_v^i \| \mathcal{P}_c\}_{c=1}^C$, where $\|$ denotes the concatenation operation. Similarly, the text template prompt tokens $\mathcal{P}_t$ are concatenated with the class tokens to obtain the text-template-prompted semantic input for C categories, denoted as $\mathcal{T}_t = \{\mathcal{P}_t \| \mathcal{P}_c\}_{c=1}^C$. A text encoder $\mathcal{G}_\varphi$ with parameters φ is then used to independently learn the category semantic representations under visual context prompts ($\mathbf{G}_v^i$) and text

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

template prompts ($\mathbf{G}_t$)

$$\mathbf{G}_v^i = \mathcal{G}_\varphi\left(\mathcal{T}_v^{\,i}\right) \in \mathbb{R}^{C\times \dim} \tag{6}$$

$$\mathbf{G}_t = \mathcal{G}_\varphi(\mathcal{T}_t) \in \mathbb{R}^{C\times \dim}. \tag{7}$$

It is worth noting that the dynamically generated visual prompt tokens, like the fixed unified text prompt tokens, supplement the description of every category label, even when the category of the current visual image is unseen. This operation provides consistent detailed information for the same category while introducing appropriate perturbations in dissimilar categories, thereby enhancing interclass discrimination.

2) Dual-Gated Adaptive Fusion: The resulting category semantic representations are dynamically integrated through a dual-gated adaptive mechanism. As illustrated in Fig. 5, the proposed A-F block employs a dual-constrained adaptive fusion strategy, which adjusts the fusion weights between fine-grained semantics derived from visual prompts and coarse-grained semantics obtained from text prompts. This adjustment is guided by the classification uncertainty and the visual features of the current image, thereby improving the accuracy and robustness of cross-scene HSIC.

Essentially, we expect the fine-grained semantics from visual prompts to capture local context and visual details. When the coarse-grained semantics from text prompts are insufficient for accurate classification, it can enhance semantic matching for the correct category while weakening the matching for incorrect categories. To achieve this, classification uncertainty is used to define an enhancement-suppression weight for the two types of semantics, dynamically adjusting their contributions for different categories. Classification uncertainty is determined by the similarity between the visual representation of the current image and the semantic representation of all categories. The enhancement-suppression weights for the semantics can be expressed as

$$\boldsymbol{\alpha}_i = \mathrm{Softmax}\left(\mathbf{f}_i \mathbf{G}_t^\top\right) \tag{8}$$

$$\boldsymbol{\beta}_i = \mathrm{Softmax}\left(\mathbf{f}_i \mathbf{G}_v^{i\top}\right). \tag{9}$$

Here $\boldsymbol{\alpha}_i \in \mathbb{R}^{1\times C}$ measures the relative importance of coarse-grained semantics for each category i among C categories, and $\boldsymbol{\beta}_i \in \mathbb{R}^{1\times C}$ represents the relative importance of fine-grained semantics, and the Softmax function normalizes the weights.

Next, a dynamic fusion coefficient γ driven by image features $\mathbf{f}_i$ is introduced. Using nonlinear weighting, it dynamically fuses the coarse-grained and fine-grained semantic representations, allowing the model to flexibly adapt to category semantic variations across different scenes, thereby improving the robustness of cross-scene classification. Therefore, the dynamically generated category semantic representation corresponding to the current visual image can be expressed as

$$\mathbf{G}_i = (1-\gamma)\odot\left(\boldsymbol{\alpha}_i^\top \odot \mathbf{G}_t\right) + \gamma \odot \left(\boldsymbol{\beta}_i^\top \odot \mathbf{G}_v^i\right) \tag{10}$$

where $\mathbf{G}_i \in \mathbb{R}^{C\times \dim}$, the dynamic fusion coefficient γ is implemented via a two-layer multilayer perceptron, $\gamma = \mathrm{MLP}(\mathbf{f}_i)$.

TABLE I
CLASSES AND NUMBERS OF SAMPLES IN HOUSTON DATASETS

Class No.	Class Name	Source (Houston 2013)	Target (Houston 2018)
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
Total		2530	53200

C. Semantic Alignment Transfer

This section maps visual representations into a shared category semantic space through cross-modal semantic alignment between visual and textual modalities. This approach maintains the stability of category semantics across scenes, aiming to mitigate the impact of domain bias on classification performance. Finally, by generating adaptive semantic weights for categories, zero-shot transfer across scenes is achieved.

1) Semantic Consistency Alignment: To align visual and textual modalities, we first compute the cosine similarity between the visual representation $\mathbf{f}_i$ of the ith hyperspectral image patch $\mathbf{X}_i \in \mathcal{X}$ and the category semantic representation $\mathbf{G}_i = \{\mathbf{g}_c^i\}_{c=1}^C$, obtaining the similarity between the ith image patch and the cth category semantic $S_{ic} = (\mathbf{f}_i^\top \mathbf{g}_c^i)/(\|\mathbf{f}_i\| \cdot \|\mathbf{g}_c^i\|)$. Based on the similarity between the current visual image patch and C category semantics, we further compute the adaptive semantic weight

$$A_{ic} = \frac{\exp(S_{ic}/\tau)}{\sum_{c'=1}^{C}\exp(S_{ic'}/\tau)}. \tag{11}$$

Here, A_{ic} reflects the association between the ith visual image patch and the cth category while also revealing the probability of the image patch being classified as category c. τ is a scaling factor introduced to stabilize the training process. Traditional hard alignment methods often assume a strict one-to-one correspondence between visual features and categories, ignoring intraclass variations, which limits the model's generalization ability. In contrast, the proposed soft alignment mechanism allows flexible matching of visual representations across categories, enhancing adaptability to intraclass variations while reducing cross-scene domain shift effects.

To optimize the alignment process, we employ a cross-entropy loss to maximize the matching weight between the visual representation and the correct category while minimizing its similarity with incorrect categories

$$L = -\frac{1}{N}\sum_{i=1}^{N}\log A_{i\,y_i} \tag{12}$$

where y_i represents the ground-truth category label for the ith image patch.

2) Zero-Shot Transfer: Our work emphasizes the cross-scene adaptability of semantic knowledge. Here, the semantic weight matrix not only serves as a flexible matching tool for known categories but also as a domain-invariant knowledge structure that can be extended to new scenes. Using the generated semantic consistency representation, the model maintains a high generalization of categories across

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

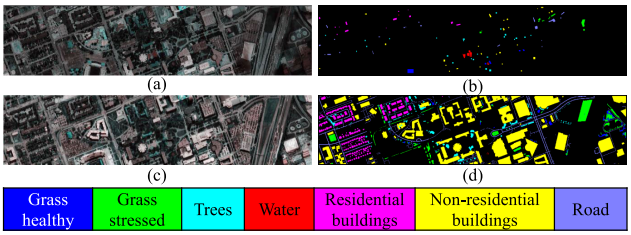


Fig. 6. Visualization of Houston datasets. (a) Houston 2013 false-color image. (b) Houston 2013 ground-truth. (c) Houston 2018 false-color image. (d) Houston 2018 ground-truth.

TABLE II
CLASSES AND NUMBERS OF SAMPLES IN PAVIA DATASETS

Class No.	Class Name	Source (PaviaU)	Target (PaviaC)
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
Total		39332	39355

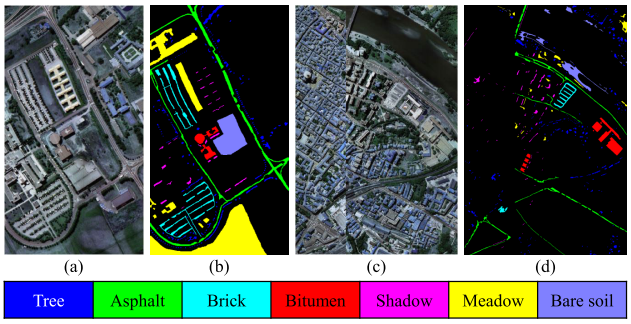


Fig. 7. Visualization of Pavia datasets. (a) PaviaU false-color image. (b) PaviaU ground-truth. (c) PaviaC false-color image. (d) PaviaC ground-truth.

scenes, possessing the capability to infer category alignment in unseen scenarios.

During inference, the hyperspectral visual image of the target scene is input to the model, which generates visual prompt semantics based on the contextual information of the current visual image. This is then fused with fixed textual template semantics to compute the semantic space of all possible categories corresponding to the current image. Finally, through similarity calculations between visual embeddings and category semantic representations, the class with the highest probability is selected as the predicted class.

IV. EXPERIMENTS

A. Experimental Datasets

1) Houston: The Houston dataset comprises scenes collected in 2013 and 2018, located in and around the University of Houston campus in the United States, captured by different sensors. Houston 2013 dataset serves as the source scene, with an image size of 349 × 1905 pixels, 144 spectral bands, and a wavelength range from 380 to 1050 nm, offering a spatial resolution of 2.5 m. Houston 2018 dataset, used as the target scene, contains 48 spectral bands within the same wavelength range and a spatial resolution of 1 m. To align scenes across the years, 48 matching spectral bands from the Houston 2013 dataset are extracted, and a 209 × 955 pixel overlapping region is selected for research. Both datasets contain seven consistent land cover categories, with specific category names and sample counts listed in Table I. Pseudo-color images and ground truth are shown in Fig. 6.

TABLE III
CLASSES AND NUMBERS OF SAMPLES IN HYRANK DATASETS

Class No.	Class Name	Source (Dioni)	Target (Loukia)
1	Dense urban fabric	1262	206
2	Mineral extraction sites	204	54
3	Non irrigated arable land	614	426
4	Fruit trees	150	79
5	Olive groves	1768	1107
6	Coniferous forest	361	422
7	Dense sderophyllous vegetation	5035	2996
8	Sparce sderophyllous vegetation	6374	2361
9	Sparcely vegetated areas	1754	399
10	Rocks and sand	492	453
11	Water	1612	1393
12	Coastal water	398	421
Total		20024	10317

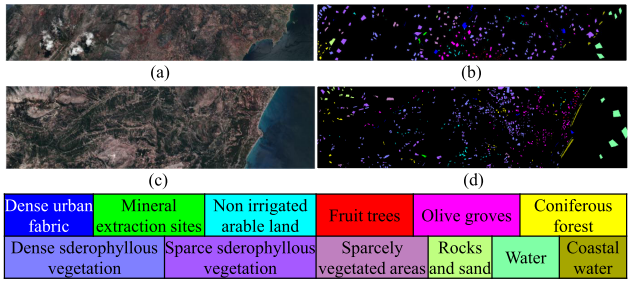


Fig. 8. Visualization of HyRANK datasets. (a) Dioni false-color image. (b) Dioni ground-truth. (c) Loukia false-color image. (d) Loukia ground-truth.

2) Pavia: The Pavia dataset comprises scenes from the University of Pavia (PaviaU) and Pavia Center (PaviaC) in northern Italy, captured by the ROSIS sensor. The PaviaU scene serves as the source scene, with a size of 610 × 340 pixels and a spatial resolution of 1.3 m. The PaviaC scene, used as the target scene, measures 1096 × 715 pixels and contains 102 spectral bands. To ensure consistency in the number of spectral bands across scenes, the last spectral band of the PaviaU scene is removed, resulting in 103 spectral bands. Both scenes contain seven consistent land cover categories, with specific category names and sample counts listed in Table II. Pseudo-color images and ground truth are displayed in Fig. 7.

3) HyRANK: The HyRANK dataset includes two scenes, Dioni and Loukia, captured by the Hyperion sensor, each with a spatial resolution of 30 m and 176 spectral bands. In this dataset, Dioni serves as the source scene with a size of 250 × 1376 pixels, while Loukia serves as the target scene with a size of 249 × 945 pixels. Both scenes contain 12 consistent land cover categories, with specific category names and sample counts listed in Table III. Pseudo-color images and ground truth are shown in Fig. 8.

B. Experimental Setting

To evaluate the proposed method, we compare it with several existing approaches, including the classic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON TARGET HOUSTON 2018 USING 5% SOURCE HOUSTON 2013

Class No.	DAN [20]	DSAN [32]	TSTNet [33]	SDENet [34]	JAN-SIF [35]	LDGNet [8]	EHSnet [36]	TASA-Net
1	$40.00_{\pm 6.41}$	$45.47_{\pm 7.27}$	$39.24_{\pm 5.26}$	$41.40_{\pm 8.60}$	$53.62_{\pm 8.19}$	$\mathbf{81.73_{\pm 6.28}}$	$31.53_{\pm 14.3}$	$61.38_{\pm 6.61}$
2	$39.84_{\pm 4.97}$	$42.06_{\pm 5.51}$	$68.95_{\pm 5.27}$	$54.42_{\pm 9.90}$	$60.77_{\pm 5.22}$	$64.94_{\pm 6.69}$	$62.89_{\pm 10.7}$	$\mathbf{70.15_{\pm 3.95}}$
3	$58.57_{\pm 5.96}$	$56.78_{\pm 12.2}$	$74.23_{\pm 13.6}$	$29.53_{\pm 15.7}$	$70.93_{\pm 6.21}$	$68.24_{\pm 7.49}$	$50.96_{\pm 6.48}$	$\mathbf{78.42_{\pm 10.1}}$
4	$3.214_{\pm 4.65}$	$26.62_{\pm 38.4}$	$3.435_{\pm 1.56}$	$6.266_{\pm 7.95}$	$6.541_{\pm 4.35}$	$\mathbf{43.24_{\pm 27.7}}$	$4.509_{\pm 5.83}$	$30.57_{\pm 35.2}$
5	$38.35_{\pm 4.95}$	$36.39_{\pm 3.60}$	$40.16_{\pm 8.53}$	$23.17_{\pm 11.9}$	$42.18_{\pm 3.26}$	$\mathbf{55.93_{\pm 10.0}}$	$37.53_{\pm 6.31}$	$50.09_{\pm 5.31}$
6	$91.95_{\pm 2.90}$	$92.66_{\pm 2.56}$	$89.41_{\pm 0.99}$	$85.37_{\pm 6.18}$	$\mathbf{95.93_{\pm 0.34}}$	$83.43_{\pm 3.75}$	$85.18_{\pm 0.58}$	$89.18_{\pm 2.56}$
7	$28.17_{\pm 4.37}$	$38.00_{\pm 8.25}$	$61.16_{\pm 13.6}$	$23.84_{\pm 8.31}$	$\mathbf{70.27_{\pm 6.50}}$	$66.43_{\pm 18.9}$	$42.67_{\pm 6.93}$	$66.79_{\pm 9.33}$
OA(%)	$56.46_{\pm 4.48}$	$61.78_{\pm 3.15}$	$70.41_{\pm 3.35}$	$61.48_{\pm 2.41}$	$74.13_{\pm 2.90}$	$75.38_{\pm 1.52}$	$67.35_{\pm 1.78}$	$\mathbf{77.35_{\pm 1.71}}$
AA(%)	$42.87_{\pm 2.65}$	$48.28_{\pm 5.42}$	$53.80_{\pm 3.30}$	$37.72_{\pm 5.99}$	$57.18_{\pm 2.07}$	$\mathbf{66.28_{\pm 5.03}}$	$45.04_{\pm 3.70}$	$63.80_{\pm 4.44}$
Kappa($\times 100$)	$39.52_{\pm 3.49}$	$44.80_{\pm 2.93}$	$53.60_{\pm 4.21}$	$38.10_{\pm 3.49}$	$61.07_{\pm 3.55}$	$56.23_{\pm 4.42}$	$46.89_{\pm 2.13}$	$\mathbf{61.96_{\pm 2.55}}$

TABLE V
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON TARGET PAVIAC USING 5% SOURCE PAVIAU

Class No.	DAN [20]	DSAN [32]	TSTNet [33]	SDENet [34]	JAN-SIF [35]	LDGNet [8]	EHSnet [36]	TASA-Net
1	$79.24_{\pm 5.80}$	$86.77_{\pm 4.28}$	$\mathbf{93.37_{\pm 0.64}}$	$87.87_{\pm 0.87}$	$90.51_{\pm 4.16}$	$86.19_{\pm 3.79}$	$87.41_{\pm 2.27}$	$92.42_{\pm 3.11}$
2	$72.51_{\pm 3.96}$	$63.97_{\pm 6.45}$	$69.76_{\pm 6.95}$	$84.39_{\pm 3.40}$	$79.77_{\pm 9.06}$	$82.00_{\pm 4.93}$	$84.50_{\pm 2.57}$	$\mathbf{89.24_{\pm 1.76}}$
3	$26.43_{\pm 5.56}$	$42.27_{\pm 8.45}$	$33.78_{\pm 8.07}$	$55.93_{\pm 3.89}$	$42.71_{\pm 8.91}$	$52.19_{\pm 4.59}$	$57.68_{\pm 7.67}$	$\mathbf{63.32_{\pm 6.27}}$
4	$88.52_{\pm 2.39}$	$87.88_{\pm 4.06}$	$71.00_{\pm 29.3}$	$87.51_{\pm 4.82}$	$\mathbf{97.95_{\pm 3.12}}$	$82.90_{\pm 4.57}$	$76.42_{\pm 5.24}$	$93.60_{\pm 1.31}$
5	$95.36_{\pm 2.65}$	$93.95_{\pm 4.76}$	$\mathbf{99.89_{\pm 0.12}}$	$92.76_{\pm 6.23}$	$99.78_{\pm 0.13}$	$94.41_{\pm 3.40}$	$82.41_{\pm 7.78}$	$98.07_{\pm 1.05}$
6	$27.23_{\pm 2.67}$	$33.24_{\pm 6.89}$	$32.82_{\pm 2.32}$	$47.01_{\pm 1.68}$	$48.81_{\pm 10.3}$	$65.48_{\pm 14.1}$	$67.27_{\pm 4.55}$	$\mathbf{71.20_{\pm 5.74}}$
7	$75.09_{\pm 5.57}$	$86.17_{\pm 6.67}$	$69.02_{\pm 5.42}$	$90.62_{\pm 3.39}$	$79.16_{\pm 5.19}$	$84.98_{\pm 9.00}$	$89.34_{\pm 5.03}$	$\mathbf{93.35_{\pm 2.62}}$
OA(%)	$63.79_{\pm 3.03}$	$66.30_{\pm 4.24}$	$61.95_{\pm 1.26}$	$79.28_{\pm 0.97}$	$74.91_{\pm 4.42}$	$80.46_{\pm 2.88}$	$80.49_{\pm 1.44}$	$\mathbf{87.43_{\pm 0.49}}$
AA(%)	$66.34_{\pm 1.04}$	$70.61_{\pm 2.84}$	$67.09_{\pm 4.43}$	$78.01_{\pm 0.41}$	$76.95_{\pm 2.53}$	$78.31_{\pm 3.16}$	$77.86_{\pm 1.62}$	$\mathbf{85.89_{\pm 0.43}}$
Kappa($\times 100$)	$56.95_{\pm 3.56}$	$59.65_{\pm 5.04}$	$55.08_{\pm 1.48}$	$75.24_{\pm 1.19}$	$70.08_{\pm 5.23}$	$76.40_{\pm 3.40}$	$76.45_{\pm 1.72}$	$\mathbf{84.92_{\pm 0.56}}$

unsupervised domain adaptation method (such as DAN [20] and DSAN [32]), and state-of-the-art (SOTA) methods primarily designed for cross-scene HSIC (such as TSTNet [33], SDENet [34], JAN-SIF [35], LDGNet [8]), and EHSnet [36]. Among these, JAN-SIF, LDGNet, and EHSnet are rapidly evolving semantic-guided methods. In the DAN and DSAN experiments, we adapt the visual encoder used in our proposed method for feature extraction and respectively employ the MMD loss and local MMD (LMMD) loss to accommodate cross-scene classification tasks. For methods specifically designed for cross-scene classification, we utilize the optimal parameter settings as reported in their original articles. In cases where parameters for specific datasets are not provided, we conduct independent parameter tuning. During the experiments, we randomly select 5% of the training set samples for training, and all samples in the testing set are used for model evaluation. Notably, in comparison experiments other than SDENet, LDGNet, EHSnet, and the proposed methods, the unlabeled data from the target scene are used for adversarial training.

For the proposed method, a two-layer 3-D residual CNN network is employed as the visual encoder. Considering computational complexity, we extract text features using the first three layers of the text encoder from the CLIP method, with the pretrained ViT-B-32.pt model used as the initialization for the text encoder. For Houston, Pavia, and HyRANK datasets, the number of attention blocks in the visual-guided fine-grained semantic prompt module is set to 2, while the number of projector blocks is set to 4, 4, and 5, respectively. The patch size of hyperspectral images for model input is respectively set to 13, 7, and 7, then "A hyperspectral image of [CLASS]." is designed as the text prompt template. Additionally, the Adam optimizer is used with 200 epochs, a batch size of 64, and learning rates of 0.01 (Houston), 0.001 (Pavia), and 0.001 (HyRANK). Finally, all experiments are repeated five times with the random setup, and their average results are used as the basis for model performance comparison, where the evaluation metrics include overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

C. Comparative Experiments and Results Analysis

In this section, we compare the proposed TASA-Net with classical unsupervised domain adaptation methods and several SOTA cross-scene HSIC models.

1) Comparison With Classical Methods: DAN aligns the global feature distributions of source and target scenes using MMD, effectively reducing domain discrepancies. However, its global MMD alignment approach overlooks class-level semantic structures, leading to suboptimal performance in target scenes with large distribution differences. DSAN, on the other hand, extends DAN by introducing a subdomain alignment mechanism, enabling finer-grained feature adaptation for different class-specific subdistributions. Nevertheless, their alignment mechanism relies on visual features to infer class distributions, which limits its ability to effectively leverage semantic information, reducing its capacity to differentiate semantically complex categories. In contrast, TASA-Net incorporates text-driven semantic guidance into the alignment process to explicitly address class-level semantic features, making it more advantageous in complex semantic scenarios.

2) Comparison With SOTA Methods: TSTNet integrates topological structures with semantic information but relies on traditional CNN features for semantic extraction, limiting its ability to fully capture global semantic relationships between classes. SDENet improves intraclass consistency through generative adversarial networks but lacks explicit semantic alignment mechanisms, leading to indirect and less precise semantic feature capture. JAN-SIF explores semantic alignment to some extent; however, it primarily derives semantic information from visual features, resulting in less flexible

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
CLASS-SPECIFIC CLASSIFICATION ACCURACY (%) ON TARGET LOUKIA USING 5% SOURCE DIONI

Class No.	DAN [20]	DSAN [32]	TSTNet [33]	SDENet [34]	JAN-SIF [35]	LDGNet [8]	EHSnet [36]	TASA-Net
1	$22.09_{\pm 6.96}$	$22.58_{\pm 4.71}$	$15.50_{\pm 4.36}$	$2.163_{\pm 1.74}$	$33.25_{\pm 19.0}$	$28.45_{\pm 37.4}$	$13.98_{\pm 19.2}$	$\mathbf{52.38_{\pm 2.93}}$
2	$12.32_{\pm 12.4}$	$44.17_{\pm 41.5}$	$26.19_{\pm 13.2}$	$44.62_{\pm 22.9}$	$26.82_{\pm 10.8}$	$32.76_{\pm 40.2}$	$14.46_{\pm 15.5}$	$\mathbf{52.31_{\pm 32.3}}$
3	$58.18_{\pm 27.2}$	$64.19_{\pm 26.0}$	$64.23_{\pm 16.8}$	$28.10_{\pm 36.4}$	$61.57_{\pm 34.4}$	$37.39_{\pm 28.4}$	$45.63_{\pm 23.4}$	$\mathbf{79.93_{\pm 8.79}}$
4	$15.00_{\pm 7.18}$	$\mathbf{33.38_{\pm 28.0}}$	$20.51_{\pm 20.9}$	$4.615_{\pm 9.23}$	$5.743_{\pm 8.44}$	$17.46_{\pm 16.2}$	$12.36_{\pm 16.9}$	$2.735_{\pm 4.30}$
5	$66.82_{\pm 6.67}$	$58.83_{\pm 7.79}$	$\mathbf{73.30_{\pm 1.61}}$	$32.97_{\pm 36.6}$	$71.29_{\pm 5.13}$	$11.08_{\pm 14.4}$	$21.27_{\pm 19.4}$	$53.64_{\pm 17.7}$
6	$\mathbf{94.07_{\pm 8.32}}$	$91.01_{\pm 11.4}$	$66.61_{\pm 12.6}$	$16.56_{\pm 28.8}$	$61.68_{\pm 38.1}$	$67.22_{\pm 34.3}$	$49.07_{\pm 35.4}$	$74.04_{\pm 10.1}$
7	$69.75_{\pm 4.32}$	$66.76_{\pm 3.33}$	$71.70_{\pm 2.28}$	$63.89_{\pm 3.32}$	$\mathbf{74.47_{\pm 4.26}}$	$55.72_{\pm 2.32}$	$59.11_{\pm 2.90}$	$69.76_{\pm 8.23}$
8	$40.48_{\pm 6.96}$	$44.02_{\pm 8.94}$	$56.00_{\pm 1.68}$	$51.74_{\pm 2.15}$	$56.43_{\pm 2.47}$	$55.02_{\pm 3.85}$	$57.00_{\pm 3.77}$	$\mathbf{58.34_{\pm 1.35}}$
9	$17.31_{\pm 2.62}$	$20.60_{\pm 1.98}$	$22.78_{\pm 1.54}$	$24.23_{\pm 7.68}$	$22.45_{\pm 2.55}$	$14.91_{\pm 5.68}$	$\mathbf{25.68_{\pm 4.18}}$	$23.86_{\pm 4.16}$
10	$0.917_{\pm 1.83}$	$00.00_{\pm 0.00}$	$44.52_{\pm 32.2}$	$28.43_{\pm 33.2}$	$33.10_{\pm 42.0}$	$3.636_{\pm 7.27}$	$41.44_{\pm 33.9}$	$\mathbf{77.74_{\pm 16.5}}$
11	$95.80_{\pm 2.63}$	$92.82_{\pm 3.63}$	$99.93_{\pm 0.06}$	$92.75_{\pm 7.81}$	$98.84_{\pm 2.33}$	$\mathbf{99.96_{\pm 0.09}}$	$97.38_{\pm 2.52}$	$99.94_{\pm 0.08}$
12	$93.57_{\pm 7.89}$	$94.38_{\pm 8.33}$	$90.38_{\pm 4.43}$	$76.90_{\pm 8.46}$	$96.59_{\pm 3.95}$	$94.05_{\pm 6.85}$	$\mathbf{99.38_{\pm 0.68}}$	$94.25_{\pm 5.04}$
OA(%)	$52.63_{\pm 3.63}$	$54.90_{\pm 4.34}$	$63.47_{\pm 1.05}$	$58.90_{\pm 1.22}$	$64.68_{\pm 1.30}$	$60.16_{\pm 0.01}$	$61.72_{\pm 0.77}$	$\mathbf{66.44_{\pm 1.76}}$
AA(%)	$48.86_{\pm 3.05}$	$52.73_{\pm 4.71}$	$54.30_{\pm 4.61}$	$38.91_{\pm 8.43}$	$53.52_{\pm 6.95}$	$43.14_{\pm 12.1}$	$44.73_{\pm 8.93}$	$\mathbf{61.58_{\pm 3.20}}$
Kappa($\times 100$)	$43.83_{\pm 3.21}$	$45.73_{\pm 4.56}$	$55.55_{\pm 1.39}$	$48.58_{\pm 1.93}$	$56.98_{\pm 1.73}$	$49.36_{\pm 1.46}$	$51.63_{\pm 1.37}$	$\mathbf{58.66_{\pm 2.66}}$

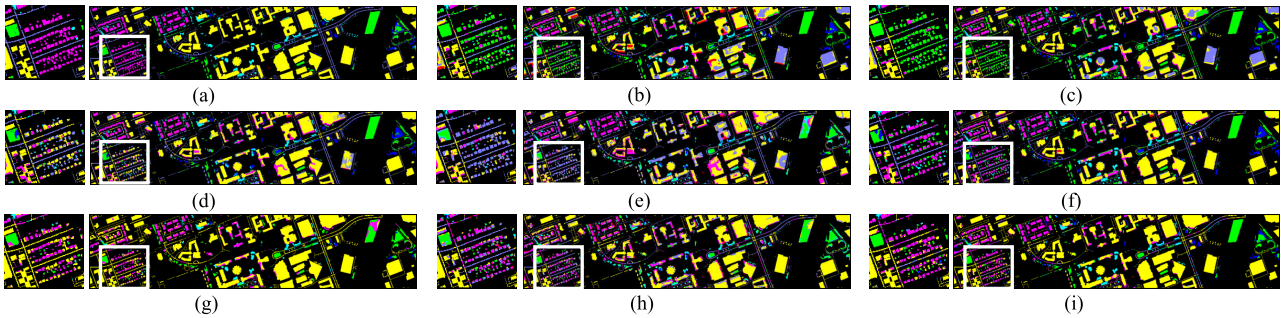


Fig. 9. (a) Ground-truth. (b)–(i) Classification maps of target Houston 2018. (b) DAN. (c) DSAN. (d) TSTNet. (e) SDENet. (f) JAN-SIF. (g) LDGNet. (h) EHSnet. (i) TASA-Net.

semantic modeling. Similarly, LDGNet introduces language modality alignment for the first time, achieving semantic sharing through vision-language alignment, while EHSNet enhances textual input with global descriptions that reduce subjective selectivity; yet, both rely on fixed text characteristics, limiting their adaptability to dynamic semantic changes in the target domain. In contrast, TASA-Net achieves more accurate cross-scene classification by employing text-driven adaptive semantic alignment and directly modeling class-level semantic relationships, while also leveraging dynamic text embedding to enrich the semantic context and enhance the robustness of cross-scene classification tasks.

3) Analysis of Classification Results: The experimental results presented in Tables IV–VI demonstrate the remarkable classification performance of TASA-Net on all the given datasets. Taking the Pavia dataset as an example, where the proposed TASA-Net achieves an OA of 87.43%, it surpasses the best-performing classical method DSAN by 21.13% and outperforms the SOTA EHSnet by 6.94%. Furthermore, the comparison of class-specific classification accuracy highlights the robustness of TASA-Net in handling transitional areas between categories. Notably, it delivers stable performance on challenging regions such as class No. 3 in the Houston dataset, class No. 2 in the Pavia dataset, and class No. 1 in the HyRANK dataset. Additionally, TASA-Net demonstrates superior capability in distinguishing semantically similar categories, as observed in the HyRANK dataset, where it effectively differentiates "sparse sderophyllous vegetation" from "dense sderophyllous vegetation." Meanwhile, the classification maps (shown in Figs. 9–11) provide an intuitive visualization of the performance differences among various methods in the target scene. Compared to other methods, which exhibit numerous misclassifications in boundary-blurred and semantically similar regions (highlighted by white boxes in the figures), TASA-Net significantly reduces errors caused by boundary ambiguity through its text-driven semantic alignment strategy, achieving higher accuracy in semantically complex areas.

Despite achieving the best OA results across all three datasets, TASA-Net still exhibits certain limitations. Experiments reveal significant accuracy drops when target-scene spectral reflectance diverges markedly from source scenes with limited training samples. On the HYRANK dataset, the "fruit tree" category, which has very few training samples, achieved only $2.735\%_{\pm 4.30\%}$ classification accuracy, often misclassified as "Dense sclerophyllous vegetation" or "Olive groves." This observation underscores the strong dependence of existing methods on labeled source scene data, leading to suboptimal generalization in low-sample scenarios. Moreover, the proposed method relies on a pretrained CLIP model, whose general semantic space exhibits a certain degree of mismatch with domain-specific semantics in remote sensing, thereby affecting the effectiveness of textual and visual modality alignment.

V. DISCUSSION

To further analyze the proposed model, this section designs a series of experiments focusing on module performance, module design, and visualization.

A. Ablation Study

In this section, we conduct a detailed analysis of the components proposed in the proposed TASA-Net through the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

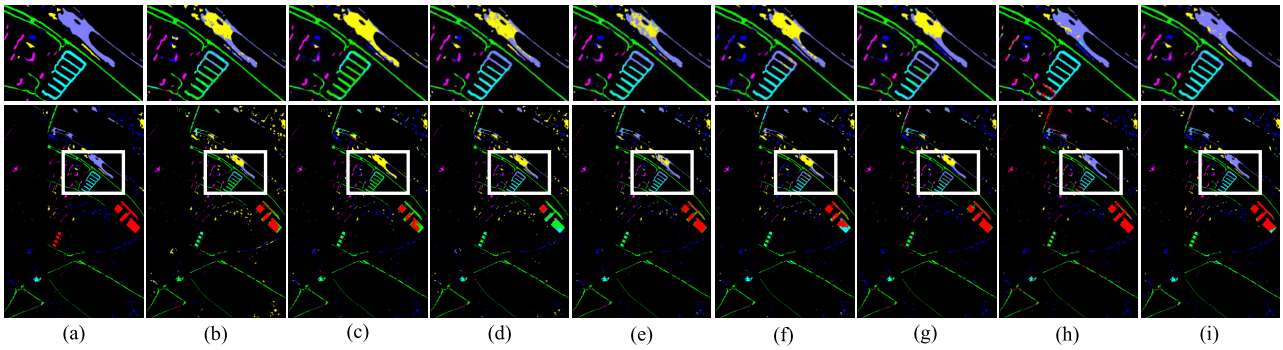


Fig. 10. (a) Ground-truth. (b)–(i) Classification maps of target PaviaC. (b) DAN. (c) DSAN. (d) TSTNet. (e) SDENet. (f) JAN-SIF. (g) LDGNet. (h) EHSnet. (i) TASA-Net.

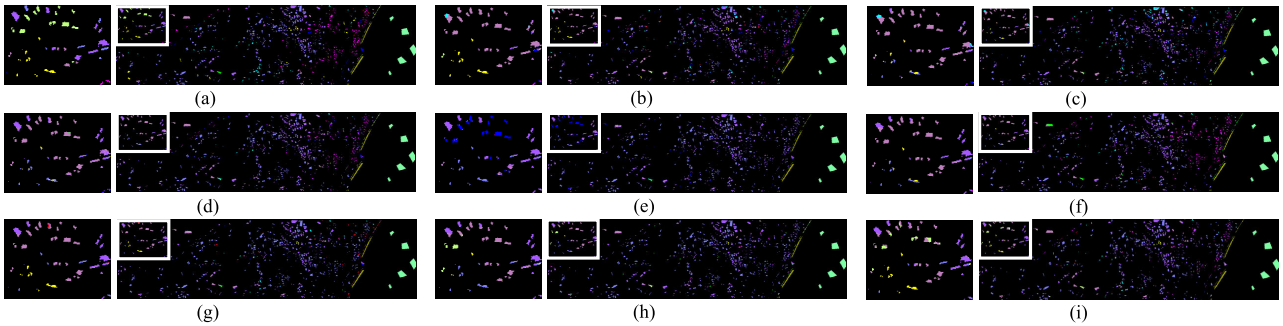


Fig. 11. (a) Ground-truth. (b)–(i) Classification maps of target Loukia. (b) DAN. (c) DSAN. (d) TSTNet. (e) SDENet. (f) JAN-SIF. (g) LDGNet. (h) EHSnet. (i) TASA-Net.

TABLE VII
CLASSIFICATION ACCURACY OF ABLATION EXPERIMENTS ON THREE DATASETS

Models	Visual inputs	Text inputs			Fusion	Accuracy	Houston	Pavia	HyRANK
		Templates prompts	Visual prompts (VG-FSP)	Labels	(DG-ASR)				
V1	✓	✗	✗	✗	✗	OA(%)	$75.21_{\pm 2.91}$	$84.48_{\pm 0.70}$	$64.37_{\pm 0.82}$
						AA(%)	$62.06_{\pm 5.32}$	$83.59_{\pm 0.70}$	$56.47_{\pm 3.23}$
						Kappa(×100)	$59.33_{\pm 2.53}$	$81.41_{\pm 0.80}$	$56.08_{\pm 1.02}$
V2	✓	✗	✗	✓	✗	OA(%)	$74.73_{\pm 2.76}$	$85.25_{\pm 1.43}$	$63.69_{\pm 1.01}$
						AA(%)	$56.98_{\pm 2.05}$	$83.56_{\pm 1.26}$	$57.16_{\pm 6.62}$
						Kappa(×100)	$59.48_{\pm 3.18}$	$82.35_{\pm 1.67}$	$54.43_{\pm 1.04}$
V3	✓	✓	✗	✓	✗	OA(%)	$75.46_{\pm 1.71}$	$85.94_{\pm 1.61}$	$64.75_{\pm 1.92}$
						AA(%)	$60.14_{\pm 3.26}$	$85.00_{\pm 0.97}$	$55.18_{\pm 2.42}$
						Kappa(×100)	$57.47_{\pm 3.37}$	$83.14_{\pm 1.91}$	$56.54_{\pm 2.50}$
V4	✓	✗	✓	✓	✗	OA(%)	$73.12_{\pm 1.87}$	$86.31_{\pm 0.68}$	$65.04_{\pm 1.89}$
						AA(%)	$62.61_{\pm 5.14}$	$85.02_{\pm 0.49}$	$57.92_{\pm 8.28}$
						Kappa(×100)	$55.80_{\pm 4.16}$	$83.62_{\pm 0.78}$	$56.27_{\pm 2.54}$
V5	✓	✓	✓	✓	✓	OA(%)	$\mathbf{77.35_{\pm 1.71}}$	$\mathbf{87.43_{\pm 0.49}}$	$\mathbf{66.44_{\pm 1.76}}$
						AA(%)	$\mathbf{63.80_{\pm 4.44}}$	$\mathbf{85.89_{\pm 0.43}}$	$\mathbf{61.58_{\pm 3.20}}$
						Kappa(×100)	$\mathbf{61.96_{\pm 2.55}}$	$\mathbf{84.92_{\pm 0.56}}$	$\mathbf{58.66_{\pm 2.66}}$

ablation experiments presented in Table VII. This series of experiments validate the positive impact of incorporating the textual modality, category prompts, and enhanced semantic representation on improving classification performance, particularly in contributing to consistent semantic understanding across different scenes.

1) V1: To evaluate the impact of the textual modality on classification performance, we remove the text input from the proposed method. In this configuration, a fully connected classification layer is applied to the visual features for category prediction, where class labels are used in a discrete form only.

2) V2: This method replaces the fully connected layer with a language–image alignment approach, where class labels are encoded into a unified semantic space in the form of textual descriptions.

3) V3: Building on V2, this approach introduces a fixed, hand-crafted class prompt template to provide a stable semantic framework for class representation.

4) V4: In this variant, the fixed, hand-crafted class prompt template in V3 is replaced with dynamically adaptive, vision-guided fine-grained contextual prompts.

5) V5: Serving as the proposed method, it incorporates a fusion module that combines coarse-grained semantics from text prompts with fine-grained semantics from visual representation.

The experimental results reveal that the variations in data distribution across different scenes significantly impact the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

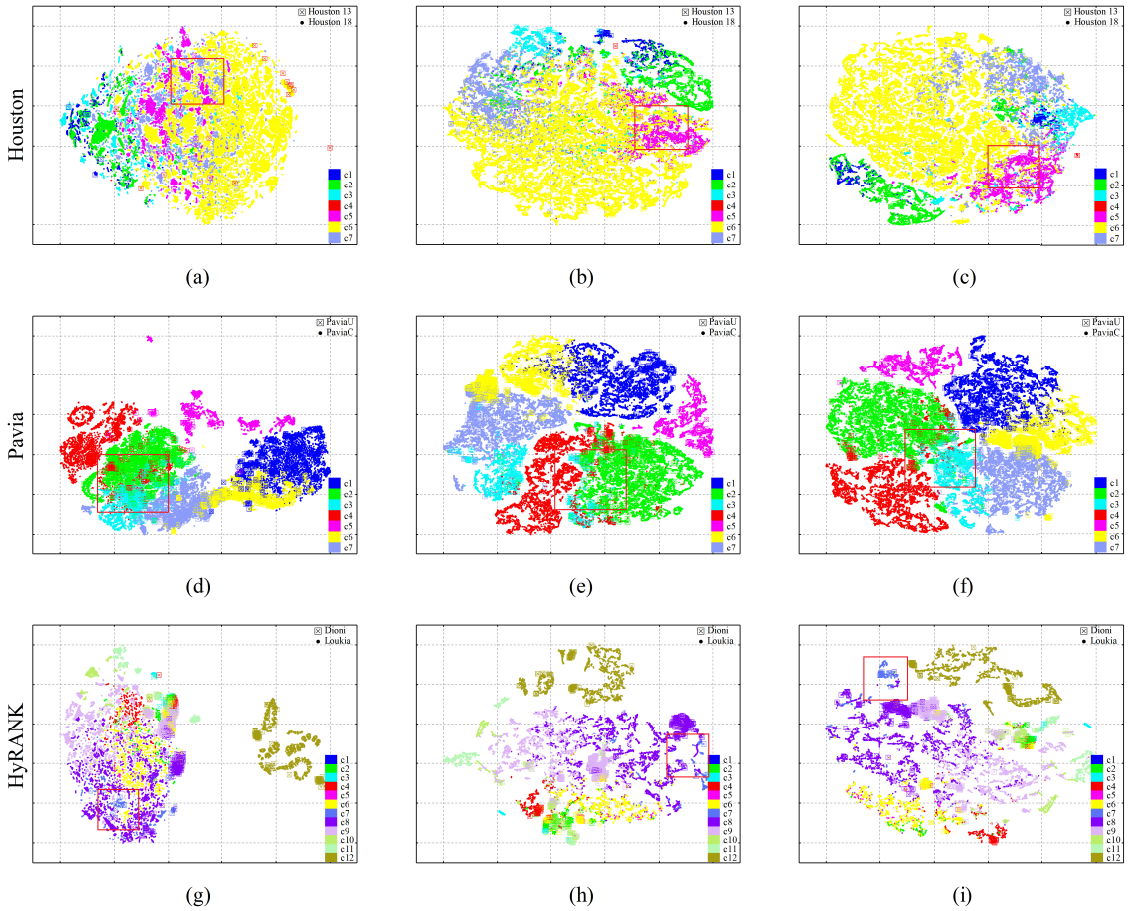


Fig. 12. T-SNE visualization of data alignment and class separability across three datasets. The figure uses the T-SNE method to visualize the feature distributions of the source and target scenes, analyzing different methods' alignment effects and class separability in cross-scene classification tasks. Each column corresponds to three different data representations. (a), (d), and (g) Original data. (b), (e), and (h) Classification based on simple category labels. (c), (f), and (i) Classification based on proposed complex semantic representation. × represents data from the source scene, while • represents data from the target scene.

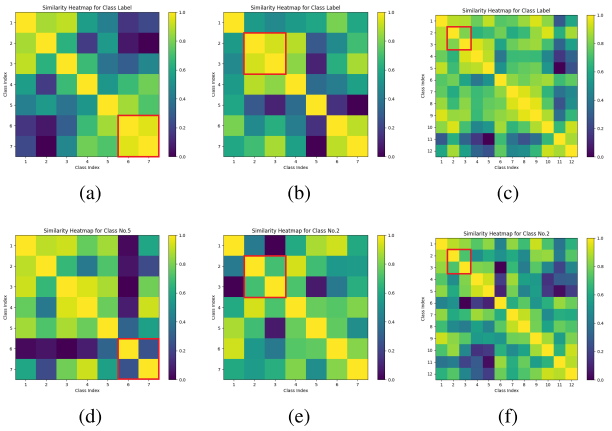


Fig. 13. Heatmaps of interclass similarity. The figure shows changes in interclass similarity under different experimental conditions to analyze the impact of visual contextual information on category discriminability. Darker colors indicate higher interclass similarity, while lighter colors indicate stronger category separability. (a)–(c) Without visual contextual information. (d)–(f) With visual contextual information. The red-boxed regions highlight areas where visual contextual information plays a crucial role.

performance of single-modal visual approaches. Specifically, the absence of semantic consistency provided by the textual modality results in a marked decline in classification performance for vision-only methods, underscoring the challenge of capturing semantic relationships between categories under scene shifts. Notably, in the V2 method, where class labels are represented as simple text inputs, the lack of a stable and generalizable semantic space leads to limited classification accuracy. However, the introduction of category prompts brings substantial performance improvements. Among these, the vision-guided fine-grained prompts (V4 method) demonstrate strong semantic representation capabilities through dynamic and adaptive integration of visual features but still fall slightly short of the proposed TASA-Net method. By integrating coarse-grained textual semantics with fine-grained, vision-guided semantics in an optimized fusion module, TASA-Net achieves further enhancements in classification accuracy and robustness across scenes. In summary, these experiments provide compelling evidence for the effectiveness of the proposed method. By refining category semantic representations within the language–image alignment framework, TASA-Net not only enhances consistent semantic understanding but also better adapts to variations in scene distributions, enabling more efficient cross-scene HSIC.

B. Visualization Analysis

The proposed method leverages semantic weighting to achieve category prediction by aligning visual features with textual semantics. Consequently, the domain invariance of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

TABLE VIII
CLASSIFICATION ACCURACY OF THE PROPOSED MODEL WITH DIFFERENT TEXT TEMPLATE PROMPTS

Text template prompts		Accuracy	Houston	Pavia	HyRANK
Parametric learned prompts		OA(%)	$74.46_{\pm 2.36}$	$84.37_{\pm 2.32}$	$64.94_{\pm 1.48}$
		AA(%)	$58.22_{\pm 3.37}$	$83.45_{\pm 1.97}$	$57.50_{\pm 9.84}$
		Kappa(×100)	$56.74_{\pm 3.73}$	$81.29_{\pm 2.75}$	$56.20_{\pm 2.28}$
Hand-crafted prompts	"*A photo of a* [CLASS]."	OA(%)	$76.10_{\pm 0.95}$	$85.07_{\pm 1.51}$	$65.35_{\pm 2.12}$
		AA(%)	$59.02_{\pm 2.32}$	$83.84_{\pm 1.09}$	$55.01_{\pm 5.30}$
		Kappa(×100)	$60.20_{\pm 2.98}$	$82.09_{\pm 1.79}$	$57.26_{\pm 2.75}$
	"*A hyperspectral image of* [CLASS]."	OA(%)	$\mathbf{77.35_{\pm 1.71}}$	$\mathbf{87.43_{\pm 0.49}}$	$\mathbf{66.44_{\pm 1.76}}$
		AA(%)	$\mathbf{63.80_{\pm 4.44}}$	$\mathbf{85.89_{\pm 0.43}}$	$\mathbf{61.58_{\pm 3.20}}$
		Kappa(×100)	$\mathbf{61.96_{\pm 2.55}}$	$\mathbf{84.92_{\pm 0.56}}$	$\mathbf{58.66_{\pm 2.66}}$

visual representations and the discriminative power of category representations are pivotal for its effectiveness. In this section, we visualize the feature distributions of the source and target domains under three different scenarios: original data, classification based on simple category labels, and classification using the proposed complex semantic representation. As shown in Fig. 12, the T-SNE method is applied to visualize the feature alignment and class separability across three datasets. The results show that the alignment effect and classification performance of the source and target scenes are significantly improved in all the scenarios, with the most pronounced enhancements observed in our approach.

To further analyze the role of visual contextual information in enhancing the discriminative power of category representations, the visualized interclass similarity heatmaps are shown in Fig. 13. The first row shows interclass classification performance without incorporating visual contextual semantics, where the results remain fixed after model training. In contrast, the second row incorporates visual contextual information from a randomly selected sample. The comparisons reveal that adding visual context significantly enhances interclass separability, even when the introduced context may not directly align with the target category semantics. As highlighted by the red boxes in Fig. 13, the inclusion of visual contextual information effectively strengthens category discriminability in complex semantic classification tasks.

C. Selection of Text Template Prompts

Several different text template prompts are designed and compared in this section to investigate how the subjectivity of prompt selection impacts classification performance. The specific approaches include the following.

1) Parametric Learned Prompts: This approach employs a parameterized method where the model automatically learns the content of a prompt with a predefined length. Through this data-driven approach, we aim to explore whether model-learned adaptive prompts perform significantly differently from manually designed templates.

2) "Photo of a [CLASS]": This template is inspired by the pretrained text encoder and aims to better match the input form accepted by the pretrained parameters. Using this template, we aim to evaluate whether aligning with the input form of the pretrained model can enhance classification performance.

3) "Hyperspectral Image of [CLASS]": This template is specifically tailored to align more closely with the characteristics of the dataset, allowing us to explore whether dataset-specific textual prompts can provide more precise semantic guidance and improve classification accuracy.

Through the exploration of different textual prompt designs, we aim to deeply understand the role of text prompts in enhancing model performance, and systematically compare the advantages and limitations of hand-crafted prompts versus parametric learned prompts in practical applications. The experimental results, as presented in Table VIII, clearly reveal the significant impact of prompt selection on model performance. While the approach of parametric learned prompts has the potential to generate adaptive prompts, its lack of explicit prior knowledge guidance leads to suboptimal classification performance in our task, highlighting the limitations of purely data-driven learning in complex scenarios. In contrast, both approaches of hand-crafted prompts exhibit superior classification performance, with the dataset-specific template achieving the best results. This finding underscores the importance of accurate and stable prior knowledge in providing clear semantic guidance to the model, effectively enhancing its robustness and generalization capability in challenging tasks. Therefore, task-specific textual prompt designs may represent a key strategy for improving model performance, offering valuable insight for further advancements in vision-language collaborative tasks.

D. Number of Visual Prompt Tokens

Visual prompts generate fine-grained semantic descriptions for categories based on the contextual information of scene images, making them closely related to the complexity of the scene. To investigate this relationship, we conduct an analysis of the number of visual contextual prompt tokens across three different datasets, aiming to understand their impact on model performance. Among them, the experiments fix the random selection of training data to reduce training time and control variables, using only 5% of the training data from a single random selection. The OA is uniformly used for the direct evaluation of the target scene data.

Fig. 14 illustrates the effect of varying the number of visual context tokens on classification performance for each dataset. The results reveal that the sensitivity of classification performance to the number of tokens varies across datasets, which we attribute to differences in the complexity of the source scenes. Specifically, as the number of tokens increases, the ability to capture fine-grained semantic representations of the proposed model improves, enhancing classification performance. However, excessive tokens may introduce semantic

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

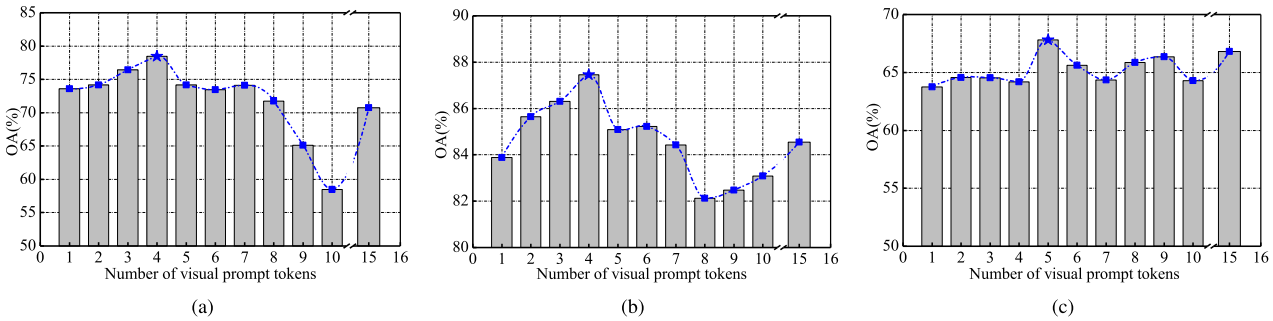


Fig. 14. Classification performance with varying numbers of visual prompt tokens on three datasets. (a) Houston. (b) Pavia. (c) HyRANK.

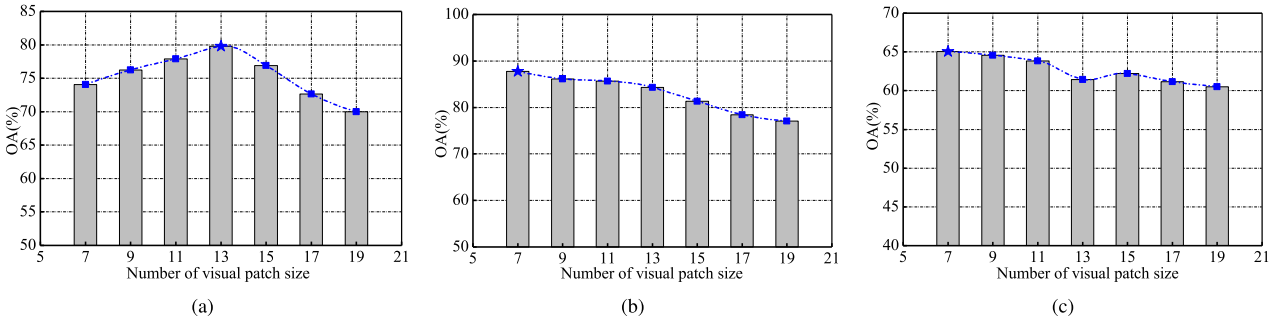


Fig. 15. Classification performance with varying numbers of visual patch size on three datasets. (a) Houston. (b) Pavia. (c) HyRANK.

redundancy, leading to diminishing returns or even a decline in performance. To address this, we carefully select the optimal number of tokens for each dataset—4, 4, and 5—balancing the trade-off between fine-grained semantic information and redundancy. These results emphasize the importance of tailoring the visual prompt configuration to the complexity of the scene, which is crucial for maximizing the model's performance while minimizing unnecessary information overhead.

E. Sensitivity Analysis of Visual Patch Size

In the proposed TASA-Net framework, the visual patch size is a critical hyperparameter affecting classification performance. An appropriately chosen patch size effectively utilizes spatial context information, thereby enhancing text-driven semantic alignment. To investigate the impact of different visual patch sizes on classification results, a sensitivity analysis is conducted with a fixed number of visual prompt tokens.

Results in Fig. 15 reveal that smaller spatial size (e.g., 7×7) in the Houston dataset fail to capture sufficient contextual information, leading to reduced classification accuracy, e.g., a 7×7 size yielded approximately 5% lower accuracy than a 13×13 size. This indicates that insufficient spatial context hampers the text-driven effects of the proposed method. However, in the Pavia and HyRANK datasets, too large patch sizes lead to performance degradation due to redundant information and noise interference. Thus, selecting an optimal patch size necessitates a trade-off between contextual richness and noise suppression, ensuring optimal classification performance.

F. Analysis of Class Semantic Representation Methods

The proposed method leverages language–image alignment to extract highly discriminative and domain-invariant

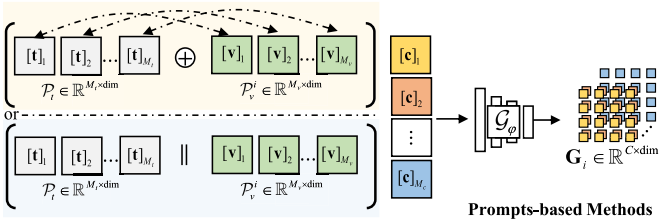


Fig. 16. Illustration of prompt-based fusion methods. The methods fuse textual template prompt tokens with visual contextual prompt tokens to generate combined prompt tokens, which are then encoded with the class labels for semantic representation. Of these, "addition" and "concatenation" are two typical combinatorial strategies.

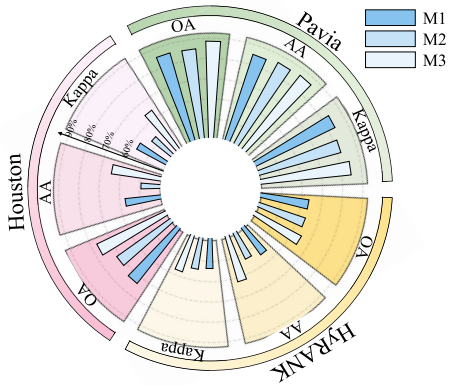


Fig. 17. Results of different fusion methods, where $M1$, $M2$, and $M3$ represent the "addition" prompt-based fusion method, "concatenation" prompt-based fusion method, and the proposed method, respectively.

representations, with the shared category semantic representations playing a crucial role in cross-scene tasks. The proposed method innovatively incorporates fine-grained semantic representations provided by visual context, improving the accuracy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

and robustness in complex scenarios. In this section, we compare the prompt-based fusion methods illustrated in Fig. 16 to integrate fine-grained semantic information from visual context and evaluate the optimal strategy for information integration. The methods fuse textual template prompt tokens with visual contextual prompt tokens to generate combined prompt tokens, which are then encoded with the class labels for semantic representation. We specifically examine two fusion strategies: "addition" and "concatenation." The addition strategy constrains the number of visual contextual prompt tokens to match the length of the textual template prompt tokens, ensuring structural consistency between the two components. In contrast, the concatenation strategy provides more flexibility, functioning as a descriptive extension of the category labels and enabling more detailed semantic expression.

The experimental results illustrated in Fig. 17 clearly demonstrate that the proposed method significantly outperforms the prompt-based methods in classification performance. This advantage stems from the feature-dependent adaptive fusion strategy of the proposed method, which dynamically balances coarse-grained and fine-grained semantics to extract class representations most suited for the current visual image classification. In contrast, the prompt-based methods lack an effective filtering mechanism for visual contextual prompt information, making it prone to incorporating irrelevant or redundant information during fusion. The introduction of such noise compromises the purity of category semantic representations, diminishing the accuracy and robustness of the model, especially in complex scenarios involving ambiguous semantics or overlapping category boundaries. These limitations further highlight the superior performance of the proposed method.

VI. Conclusion

This article presents the method called TASA-Net for cross-scene HSIC, underscoring the critical importance of textual modality in addressing cross-scene challenges. By integrating hand-crafted template prompts and VG-FSPs, TASA-Net dynamically adapts both coarse-grained and fine-grained class semantics, achieving robust and adaptive semantic fusion. Furthermore, the model aligns visual and textual modalities within a shared semantic space, effectively enabling zero-shot transfer classification across scenes. Extensive experiments on multiple cross-scene hyperspectral datasets demonstrate that TASA-Net significantly outperforms existing methods, particularly in resolving boundary ambiguities and classifying semantically similar categories. However, TASA-Net still faces limitations in effectively handling small sample categories, highlighting the need for further exploration of techniques to improve performance in such scenarios. Additionally, the proposed method constructs a semantic space based on a closed set of categories, limiting its adaptability to open-world scenarios and its ability to recognize unseen categories effectively. Future research could explore more specialized remote sensing vision-language models to enhance cross-scene classification while developing robust open-world learning strategies to improve generalization performance and application flexibility.

References

[1] X. Zheng, X. Xiao, and X. Chen, "Advancements in cross-domain remote sensing scene interpretation," *J. Image Graph.*, vol. 29, no. 6, pp. 1730–1746, Jun. 2024.

[2] X. Gao et al., "Multimodal visual-semantic representations learning for scene text recognition," *ACM Trans. Multimedia Comput., Commun., Appl.*, vol. 20, no. 7, pp. 1–18, Jul. 2024.

[3] F. Wang, X. Zhu, X. Liu, Y. Zhang, and Y. Li, "Scene graph-aware hierarchical fusion network for remote sensing image retrieval with text feedback," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4705116.

[4] L. Wang, S. Dong, Y. Chen, X. Meng, S. Fang, and S. Fei, "MetaSeg-Net: Metadata-collaborative vision-language representation learning for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5644211.

[5] Y. Bazi, M. M. A. Rahhal, M. L. Mekhalfi, M. A. A. Zuair, and F. Melgani, "Bi-modal transformer-based approach for visual question answering in remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4708011.

[6] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[7] H. Tang, W. Zhao, G. Hu, Y. Xiao, Y. Li, and H. Wang, "Text-guided diverse image synthesis for long-tailed remote sensing object classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5630413.

[8] Y. Zhang, M. Zhang, W. Li, S. Wang, and R. Tao, "Language-aware domain generalization network for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5501312.

[9] M. Singha, A. Jha, B. Solanki, S. Bose, and B. Banerjee, "APPLeNet: Visual attention parameterized prompt learning for few-shot remote sensing image generalization using CLIP," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPRW) Workshops*, Vancouver, BC, Canada, Jun. 2023, pp. 2024–2034.

[10] B. He, J. Feng, D. Li, R. Shang, J. Wu, and L. Jiao, "Learnable prompts-based transformers for domain generalization of hyperspectral image classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 10441–10445.

[11] D. Hong et al., "More diverse means better: Multimodal deep learning meets remote-sensing imagery classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4340–4354, May 2021.

[12] H. Chan-To-Hing and B. Veeravalli, "FUS-MAE: A cross-attention-based data fusion approach for masked autoencoders in remote sensing," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 6953–6958.

[13] L. Meng, J. Wang, R. Meng, Y. Yang, and L. Xiao, "A multiscale grouping transformer with CLIP latents for remote sensing image captioning," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703515.

[14] Z. Huang, H. Zhu, Y. Sun, D. Choi, C. Tan, and J.-H. Lim, "A diagnostic study of visual question answering with analogical reasoning," in *Proc. IEEE Int. Conf. Image Process. (ICIP)*, Anchorage, AK, USA, Sep. 2021, pp. 2463–2467.

[15] M. Cai et al., "Diffusion-geo: A two-stage controllable text-to-image generative model for remote sensing scenarios," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Athens, Greece, Jul. 2024, pp. 7003–7006.

[16] Y. Zhang, Q. Cheng, Z. Zhu, J. Jiao, S. Qu, and F. Zhang, "Remote sensing image detection method based on contrastive learning," in *Proc. 2nd Int. Conf. Artif. Intell. Intell. Inf. Process. (AIIIP)*, Hangzhou, China, Oct. 2023, pp. 119–123.

[17] G. Liang, P. Shivakumara, T. Lu, and C. L. Tan, "Multi-spectral fusion based approach for arbitrarily oriented scene text detection in video images," *IEEE Trans. Image Process.*, vol. 24, no. 11, pp. 4488–4501, Nov. 2015.

[18] F. Liu et al., "RemoteCLIP: A vision language foundation model for remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5622216.

[19] J. Zhu, B. Bolsterlee, Y. Song, and E. Meijering, "Improving cross-domain generalizability of medical image segmentation using uncertainty and shape-aware continual test-time domain adaptation," *Med. Image Anal.*, vol. 101, Apr. 2025, Art. no. 103422.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.

[20] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning transferable features with deep adaptation networks," in *Proc. 32nd Int. Conf. Mach. Learn.*, vol. 37, Lille, France, Jul. 2015, pp. 97–105.
[21] Y. Bi et al., "Single-source cross-domain bearing fault diagnosis via multipseudo-domain-augmented adversarial domain-invariant learning," *IEEE Internet Things J.*, vol. 11, no. 19, pp. 31521–31533, Oct. 2024.
[22] H. Chen, S. Ding, and A. Wang, "Cross-scene hyperspectral image classification based on graph alignment and distribution alignment," *Electronics*, vol. 13, no. 9, p. 1731, May 2024.
[23] X. Wang, J. Liu, Y. Ni, W. Chi, and Y. Fu, "Two-stage domain alignment single-source domain generalization network for cross-scene hyperspectral images classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5527314.
[24] S. Zhu, C. Wu, B. Du, and L. Zhang, "Robust remote sensing image cross-scene classification under noisy environment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5600411.
[25] J. Peng, Y. Huang, W. Sun, N. Chen, Y. Ning, and Q. Du, "Domain adaptation in remote sensing image classification: A survey," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 9842–9859, 2022.
[26] Y. Yang, Y. Xu, Z. Wu, B. Wang, and Z. Wei, "Cross-scene classification of hyperspectral images via generative adversarial network in latent space," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5526217.
[27] Y. Dang, X. Zhang, Y. Wang, and B. Liu, "LIVEnet: Linguistic-interact-with-visual engager domain generalization for cross-scene hyperspectral imagery classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 5506805.
[28] J. Zhou et al., "DiffDet4SAR: Diffusion-based aircraft target detection network for SAR images," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, 2024, Art. no. 4007905.
[29] C. Xiao et al., "Highly efficient and unsupervised framework for moving object detection in satellite videos," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 12, pp. 11532–11539, Dec. 2024.
[30] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. 38th Int. Conf. Mach. Learn.*, Jul. 2021, pp. 8748–8763.
[31] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. 39th Int. Conf. Mach. Learn.*, Baltimore, MD, USA, 2022, pp. 12888–12900.
[32] Y. Zhu et al., "Deep subdomain adaptation network for image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 4, pp. 1713–1722, Apr. 2021.
[33] Y. Zhang, W. Li, M. Zhang, Y. Qu, R. Tao, and H. Qi, "Topological structure and semantic information transfer network for cross-scene hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 6, pp. 2817–2830, Jun. 2023.
[34] Y. Zhang, W. Li, W. Sun, R. Tao, and Q. Du, "Single-source domain expansion network for cross-scene hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 32, pp. 1498–1512, 2023.
[35] R. Shang, Y. Xie, W. Zhang, J. Feng, and S. Xu, "Joint adversarial network with semantic and topology fusion for cross-scene hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, Jun. 2024, Art. no. 5521116.
[36] X. Wang, S. Dong, X. Zheng, R. Lu, and J. Jia, "Explicit high-level semantic network for domain generalization in hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5538314.



Wenzhen Wang received the M.S. degree from the School of Electric and Electronic Engineering, Shanghai University of Engineering Science, Shanghai, China, in 2021. She is currently pursuing the Ph.D. degree with the School of Computer Science and Engineering, Nanjing University of Science and Technology, Nanjing, China.

Her research interests include deep learning and remote sensing image processing.



Fang Liu (Member, IEEE) received the B.S. degree in information and computing science from Henan University, Kaifeng, China, in 2012, and the Ph.D. degree in intelligent information processing from Xidian University, Xi'an, China, in 2018.

She is currently an Associate Professor with Nanjing University of Science and Technology, Nanjing, China. She is also with the State Key Laboratory of Information Engineering in Surveying, Mapping and Remote Sensing, Wuhan University, Wuhan, China, and Jiangsu Province Engineering Research Center of Airborne Detecting and Intelligent Perceptive Technology, Nanjing. Her research interests include deep learning, object detection, polarimetric SAR image classification, and change detection.



Hongyuan Zhu (Member, IEEE) received the Ph.D. degree from Nanyang Technological University (NTU), Singapore, in 2015.

He is currently a Senior Scientist and PI with the Institute for Infocomm Research (I^2R), Agency for Science, Technology, and Research (A*STAR), Singapore. He is leading the Advanced Perception and Reasoning Laboratory. He has published around 90 top-tier journal and conference papers, such as CVPR, ICCV, NeurIPS, ICML, AAAI, IJCAI, ACL, and IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE. His research interests include developing autonomous agents with super large-scale multimodal sensing and reasoning capabilities for a series of use cases in manufacturing, cybersecurity, and defense with a total funding of 30 Mil $SGD.

Dr. Zhu was a recipient of A*STAR Career Award in 2022. His team won the 1st place winner of Scan2Cap challenge, ICCV2023, 3rd place in EPIC-Kitchen Text-Video Retrieval, and the only Asia team as 1st Prize Finalist of KUKA Innovation Challenge 2021. He also served as the area Chair of ACM MM Asia 2022, a Senior Program Committee of IJCAI 2021, and a Guest Editor of *IET Image Processing* in 2018. He has been an Associate Editor of *Visual Computer* since 2020. He was selected as the Top 2% scientist by Stanford in 2023



Liang Xiao (Senior Member, IEEE) received the B.S. degree in applied mathematics and the Ph.D. degree in computer science from Nanjing University of Science and Technology (NJUST), Nanjing, China, in 1999 and 2004, respectively.

From 2009 to 2010, he was a Post-Doctoral Fellow with the Rensselaer Polytechnic Institute, Troy, NY, USA. Since 2014, he has been the Deputy Director of Jiangsu Key Laboratory of Spectral Imaging Intelligent Perception, Nanjing. He has served as the Second Director of the Key Laboratory of Intelligent Perception and Systems for High-Dimensional Information of Ministry of Education, NJUST, where he is currently a Professor with the School of Computer Science. He has published more than 200 international journal articles including IEEE TRANSACTIONS ON IMAGE PROCESSING, IEEE TRANSACTIONS ON NEURAL NETWORKS AND LEARNING SYSTEMS, IEEE TRANSACTIONS ON MULTIMEDIA, IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING, IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY, and IEEE JOURNAL OF SELECTED TOPICS IN APPLIED EARTH OBSERVATIONS AND REMOTE SENSING. His main research interests include inverse problems in image processing, computer vision and image understanding, pattern recognition, and remote sensing.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 02:53:42 UTC from IEEE Xplore. Restrictions apply.